

Initial Request



user-issued



env-triggered

$$H_t = (u_1, a_1, \dots, u_t)$$

Interaction history up to turn t

Evaluated Agent

Produce response a_t



memory



tool use



workspace



act and clarify



User Agent

Observe a_t , generate u_{t+1}



persona



hidden
intents



observe
progress



response



Hidden Intents

$$I = \{i_1, \dots, i_m\}$$

Latent requirements not
stated in the initial request

completed

Resolved by the
agent without
explicit user
statement



inferred

Agent asks a
focused question;
user reveals the
intent



provided

User proactively
provides the
missing
requirement

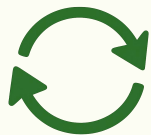


$$I = I_{\text{completed}} \cup I_{\text{inferred}} \cup I_{\text{provided}}$$

Session Termination



Terminate when every intent has a terminal status.



Turn-based
Interaction Loop

Next User Message

u_{t+1}